

3GPP

The standards organisation behind 3G, 4G and 5G technology. It's a larger body that unified several small wireless telecommunication standards development organisations, namely, ARIB, ATIS, CCSA, ETSI, TSDSI, TTA and TTC.

will come out mid-2020 is expected to have a more refined 5G standard.

Closer to home, all announcements that have taken place so far have been pre-5G technologies. Essentially, pre-5G is an enhancement of 4G with some of the technologies of 5G being clubbed with existing 4G technology. We aren't going to see any 5G deployment any time soon. Abhay Karandikar, Director, IIT Kanpur and Chairman of the recently set up committee to look into 5G spectrum told the Economic Times that, "5G trials may happen towards the end of this year or early next year. Early deployments may happen in the second or third quarter of 2020." If early deployments are to take place in the third quarter of 2020, then it stands to reason that mass deployment will happen much later. The Indian Government has directed major telecom service provider to tie up with 5G hardware manufacturers such as Cisco, Samsung, Nokia and Ericsson to run test runs this year.

KEY 5G BENEFITS

5G deployment is going to happen over multiple phases, the first of which will be focused on one of the

largest usage scenarios, eMBB. It will then be followed by more industrial use cases and so on.

eMBB

With the first wave of 5G deployment, the focus will be on eMBB (enhanced mobile broadband). The name indicates a general improvement in all cellular characteristics. This means you get to experience much higher data bandwidth, lower latency and each base station will be able to handle a lot more clients. Speeds with 5G should eventually hit 10 Gbps which is light years ahead of the 100 Mbps that 4G is supposed to provide. Since you must be very familiar with how disappointing 4G speeds are without providers, you can expect something similar when 5G is deployed. As for latency, 5G can technically bring it down to sub-10 ms and even lower if possible. Lastly, each 5G base station will be able to handle more concurrent users than what 4G is capable of. We don't have metrics from real world tests since the 5G deployment hasn't occurred yet.

mMTC

The Internet of Things is one of the largest purported usage scenario for 5G. Since 5G can handle much more concurrent traffic, it will be powerful growth accelerant for IoT adoption. mMTC stands for Massive Machine Type Communications which is a machine-centric communication and will be increasingly important as we incorporate more and more connected devices into our lives. The bulk of mMTC applications will be industrial to begin with and will continue to scale

even as the consumer applications grow. With IoT, you're looking at a lot of upstream readings from the many sensors deployed in a project. This is quite different from human centric applications which have a mix of upstream as well as downstream communication with the latter forming the bulk of it all. The 5G standard can support about 200,000 sensors in a 1 million square kilometer area.

URLLC

You'd have noticed that 5G has often been associated with certain upcoming technologies. The underlying factor that's common for all of these technologies is that they require low latency connections. For example, Autonomous Vehicles have to send massive

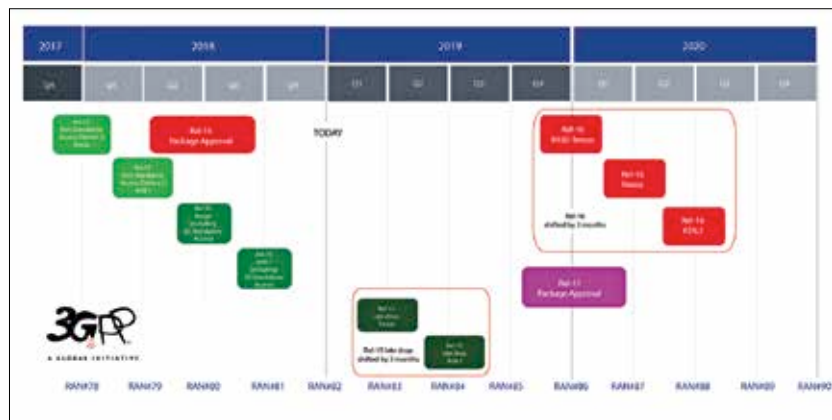


URLLC will be key to autonomous vehicles

amounts of data for processing and the vehicle's decision depends upon the timely response from the data centre. We're literally talking about life and death situations here. With 5G, the network responsiveness is expected to improve to the extent that latency will be well below 10 ms. Moreover, the error rate is expected to drastically reduce. Which is why URLLC (Ultra Reliable Low Latency Communication) is key. It's called "Ultra Reliable" because the transmission error will be lower than 1 pack loss in about 100,000 packets.

TIP OF THE ICEBERG

This was just a brief overview of the underlying technologies that are powering 5G. Obviously there's a lot more to cover and we've done so in the stories that follow. We've even gone ahead and explored what's expected of 6G and the key IoT applications that will drive the adoption of 5G all over the world. 



3GPP release schedule